Chapter 7, Problem 30P

Problem

Evaluate the following integrals involving the impulse functions:

(a) $\int_{-\infty}^{\infty} 4t^2 \delta(t-1) dt$

(b) $\int_{-\infty}^{\infty} 4t^2 \cos 2\pi t \delta(t-0.5) dt$

Step-by-step solution

Step 1 of 2

(A) $\int_{-\infty}^{\infty} 4t^2 \ \delta(t-1) \ dt$

$= \quad 4\,t^2 \ \big|_{\,t\,=\,1}$

$= \quad 4\,(1)^2$

= 4

Step 2 of 2

(B) $\int_{-\infty}^{\infty} 4t^2\, Cos\, 2\Pi t \ \ \delta(t-0.5)\, dt$

$= \quad 4t^2\, Cos\, 2\Pi t \ \big|_{\,t\,=\,0.5}$

$= \quad 4(0.5)^2\, Cos\,(2\Pi \times 0.5)$

= -1